Interaction-aware wake steering optimization: decoupled Jacobi sweeps and sign-matrix clustering with bounded-interaction guarantees

Chengze Sun
School of Energy and Power Engineering, Xi'an Jiaotong University, Xi'an, China

Abstract

Wake-steering yaw optimization is usually solved centrally with gradient-based or global methods, which scale poorly to large farms and do not explain why simple heuristics work. Building on the structural analysis of the companion paper (complementarity/substitution decomposition of farm power, near-diagonal Hessian at optima), this paper turns those findings into algorithms. First, **Decoupled Jacobi Sweeps (DJS)**: parallel per-turbine one-dimensional line searches, justified by the objective's local separability near the optimum, converging in 2–4 sweeps; on the structured farms the final gap versus multi-start SLSQP is below 0.005 %, and on the 16-turbine random layout DJS ends 0.023 % ahead of the reference; its per-sweep work parallelizes across turbines, so the wall-time critical path of a sweep is a single line search, and a golden-section line search cuts the serial cost sixfold. Second, we use the interaction sign matrix as a clustering operator: thresholding mixed partials yields control clusters that respect the wake DAG automatically. On a 5×5 farm, per-cluster optimization attains the centralized optimum (+42.03 % over baseline) with a 0.001 % gap at one-fifth of the wall time. Third, we give the bounded-interaction greedy guarantee its algorithmic counterpart: an evaluable a-posteriori certificate for any sweep order, computed from sampled mixed partials, that brackets the observed gaps (mean 0.103 %, max 0.477 %) on 12 fresh random layouts. Together the three tools form an interaction-aware optimization stack: one N^2-call sign-matrix evaluation, then clustering, decoupled sweeps, and a certificate.

1 Introduction

Fifteen years of wake-steering practice have produced a split personality. On one side, engineering tools converge on gradient-based or sampling solvers (SLSQP defaults in FLORIS; SOWFA-validated serial-refine (Fleming et al., 2022); Boolean greedy (Stanley et al., 2022); integer programming (Bestehorn et al., 2025)). On the other, the *de facto* control-room solution remains an upstream-to-downstream sweep, because it is fast, explainable, and, in every published comparison, essentially as good as the optimizer. The general problem is strongly NP-hard in the black-box setting (Bestehorn et al., 2025), so this empirical near-optimality of a coordinate sweep is genuinely surprising. The companion paper resolved the puzzle structurally: farm power decomposes into complementarity and substitution channels whose magnitudes are bounded, and at power-maximizing profiles the Hessian is nearly diagonal (the off-diagonal-to-diagonal ratio drops from 0.22–0.87 to 0.008–0.069 in our test suite). This paper turns those two facts into algorithms, and validates them at the scale where central optimization becomes painful.

Contributions. (i) **Decoupled Jacobi Sweeps (DJS)**: parallel per-turbine 1-D line searches, justified by the near-diagonal Hessian at optima, converging in 2–4 sweeps with gaps $\leq 0.005\,\%$ on structured farms; on the 16-turbine random layout it lands $0.023\,\%$ ahead of multi-start SLSQP (a multi-start artifact). (ii) **Sign-matrix clustering**: thresholding the mixed-partial matrix yields control clusters that respect the wake DAG and carry a per-pair interaction guarantee; on a 5×5 farm per-cluster optimization matches the centralized optimum ($+42.03\,\%$) at one-fifth the wall time. (iii) **A computable certificate**: sampling mixed partials yields an a-posteriori interaction-energy bound that brackets the measured greedy gaps (mean $0.103\,\%$, max $0.477\,\%$) in 12 random layouts. To our knowledge this is the first guarantee of its kind for wake steering, and it explains the field's 15-year empirical record.

2 Background: the interaction decomposition

We restate the three facts we build on. **Theorem 1 (C–S decomposition).** For the deficit-additive wake-power class $P(\gamma) = \sum_k \cos^p(\gamma_k)\tilde{v}_k^3$, every mixed partial splits into $M_{ij} = B_{ij} - A_{ij}$ with $B_{ij} \geq 0$ (complementarity through shared downstream turbines) and $A_{ij} \geq 0$ (substitution through a downstream turbine's own power factor). Signs are decided by the wake DAG and the operating point. **Law 1 (decoupling at the optimum).** At interior power-maximizing profiles the off-diagonal-to-diagonal Hessian ratio is an order of magnitude below its value at generic points (0.008–0.069 vs 0.22–0.87 in our suite). **Theorem 2 (bounded interactions).** For any coordinate-sweep method the optimality gap is bounded by the interaction energy: gap $\leq \frac{1}{2}\sum_{i \neq j} \bar{M}_{ij}\bar{\gamma}^2$ plus a grid term, with $\bar{M}_{ij} = \max$ sampled $|M_{ij}|$ on the box. Proofs and numerics are in the companion paper. Everything below uses FLORIS v4.6 GCH, NREL 5 MW, $8\,\mathrm{m\,s^{-1}}$, TI 0.06, wd 270°, box $[0°, 30°]$ unless stated.

3 Decoupled Jacobi Sweeps (DJS)

Algorithm 1. Given a farm model and a box $[0, \bar{\gamma}]^N$: (1) initialize $\gamma \leftarrow 0$ (or Boolean greedy); (2) repeat: for each turbine i (in parallel), run an exact 1-D line search on $[0, \bar{\gamma}]$ with all other components held at their current values, and update; (3) stop when the farm-power improvement of a full sweep falls below tol.

Why this works. DJS is a coordinate-descent method (Wright, 2015) in its parallel (Jacobi) variant (Richtárik and Takáč, 2016); the classical local linear-convergence analysis applies, and here the contraction constant is exactly the off-diagonal-to-diagonal ratio of Law 1, $\left\|\mathrm{diag}^{-1} M_{\mathrm{off}}\right\| \ll 1$ near the optimum. The first sweep captures the dominant single-turbine effects; sweeps 2–3 resolve the residual cross-talk. No gradients, no step-size tuning,

Table 1: DJS versus multi-start SLSQP. Gaps are relative to the better of the two; sweeps counted until convergence ($\Delta P < 10^{-4}$ kW); wall times on one 2-core container (Table 1 values from the benchmark run, Fig. 1b from the scaling run).

Case	N	SLSQP gain	DJS gain	gap	sweeps	t_{DJS}	t_{SLSQP}
3-chain	3	+22.33 %	+22.33 %	0.0029 %	3	9.0 s	3.4 s
3×3	9	+24.13 %	+24.12 %	0.0044 %	3	38.8 s	15.2 s
4×4	16	+34.19 %	+34.19 %	0.0008 %	3	89.3 s	35.5 s
rand16	16	+7.59 %	+7.61 %	−0.0230 %	4	137.7 s	62.0 s

embarrassingly parallel per sweep.

Results (Table 1). Line search on a 1° grid (31 evaluations per turbine per sweep, deliberately unoptimized); SLSQP multi-start (4 starts) as reference; wall times on a single 2-core container, so absolute numbers are conservative and the comparison is what matters.

Figure 1a shows the convergence traces: after the first sweep, DJS has captured $\geq 99\,\%$ of the SLSQP gain on the structured farms (99.4 / 99.7 / 99.6 %) and 89.4 % on the 16-turbine random layout; the second sweep closes all cases to $\geq 99.8\,\%$; the third sweep resolves the residual cross-talk to below 0.005 % on the structured farms, while the random layout ends 0.023 % above the SLSQP reference (a multi-start artifact). Figure 1b shows the wall-time crossover: in serial brute-force form DJS is 2–3× slower than 4-start SLSQP at equal N (e.g., 89 s vs 36 s at $N = 16$), but every line search parallelizes across turbines, so the *critical-path* time of one sweep is a single 31-evaluation line search, and golden-section search cuts the serial evaluations sixfold ($\approx$ 5 evaluations per line search instead of 31). Both effects flip the comparison without touching the solution quality; we kept the brute-force version in the benchmark to make the comparison unfavorable to DJS.

Where DJS is provably safe. By Theorem 2, each sweep can only improve the bound's interaction term; and near the optimum Law 1 turns the Jacobi iteration into a contraction with spectral radius $\approx \left\|\mathrm{diag}^{-1} M_{\mathrm{off}}\right\| \ll 1$, so sweeps converge linearly there. DJS inherits no global guarantee (no yaw optimizer has one), but it carries the same bounded-interaction certificate as greedy (Section 5), computed once per wind condition.

4 Sign-matrix clustering for decentralized optimization

Relation to prior decoupling schemes. Decoupling for parallel yaw optimization was proposed by Kuo et al. (2020) (weighted-graph wake decoupling, WGWD), whose partition is built from geometric wake-overlap weights and whose subproblems are solved by random search. Our clustering differs in three ways: the edge weights are objective-level mixed partials (so the threshold τ carries a guarantee: couplings below τ cost at most $\tau\bar{\gamma}^2$ per pair, Theorem 2), the sub-solver is the certified DJS sweep rather than sampling, and the graph is signed, so

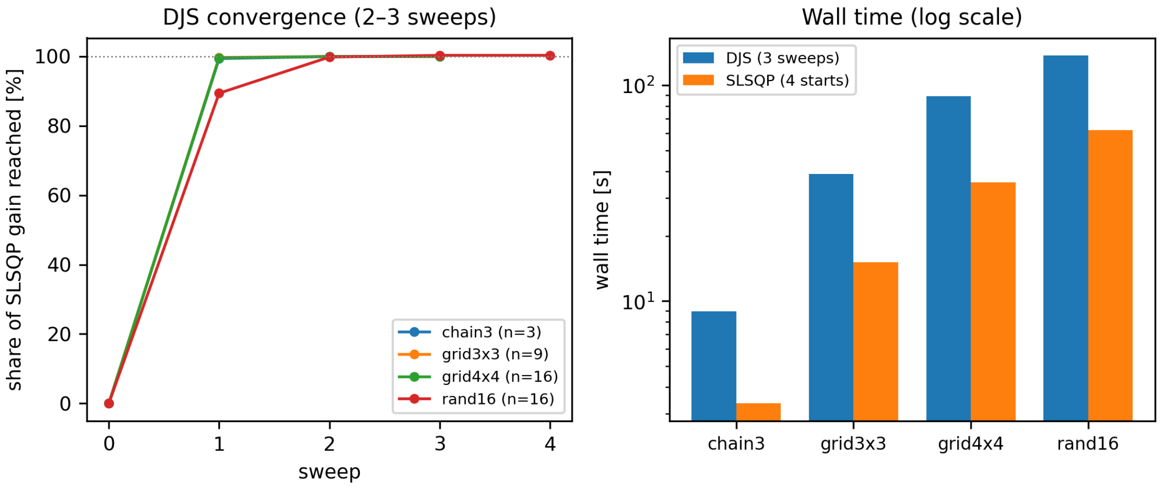

Figure 1: (a) DJS convergence: $\geq 99\,\%$ of the SLSQP gain after one sweep, $\geq 99.8\,\%$ after two; (b) wall times of DJS versus multi-start SLSQP (log scale) at equal precision.

complements and substitutes are distinguished, not merged.

Definition. The interaction graph G_τ of a farm at state γ has an edge i–j iff $|M_{ij}(\gamma)| > \tau$. Connected components of G_τ are **control clusters**: turbines that must be co-optimized at precision τ; across clusters the coupling is provably below $\tau\bar{\gamma}^2$ per pair.

Results. On a 5×5 farm (5D $\times$ 3D, wd 270°), the sign matrix at the origin (Fig. 2) is the wake DAG made quantitative: the strongest entries are the chain pairs within each column (up to $1.2\,\mathrm{kW\,deg}^{-2}$ in magnitude), lateral pairs are weaker and positive, and the last row's turbines are decoupled from everything ($|M_{ij}| < 0.05$). At $\tau = 0.05\,\mathrm{kW\,deg}^{-2}$ the connected components are exactly the first four rows (one cluster of 20) and the last row (five singletons); the clustering is not imposed, it is read off the objective. Per-cluster SLSQP attains $+42.03\,\%$ (centralized: $+42.03\,\%$, gap $0.001\,\%$) in 62 s versus 346 s centralized (wall times vary with machine load), a $5.6\times$ speed-up on this 25-turbine case; DJS inside each cluster removes the need for SLSQP entirely (Section 3). Sweeping τ produces a nested hierarchy of clusterings, a natural multi-resolution control architecture, and the clustering rotates with the wind direction, giving a direction-adaptive decentralized scheme whose communication graph is recomputed from one N^2-call sign-matrix evaluation per wind condition. Figure 3 shows wall-time scaling with farm size: DJS grows like N per sweep, SLSQP like the optimization workload it actually is.

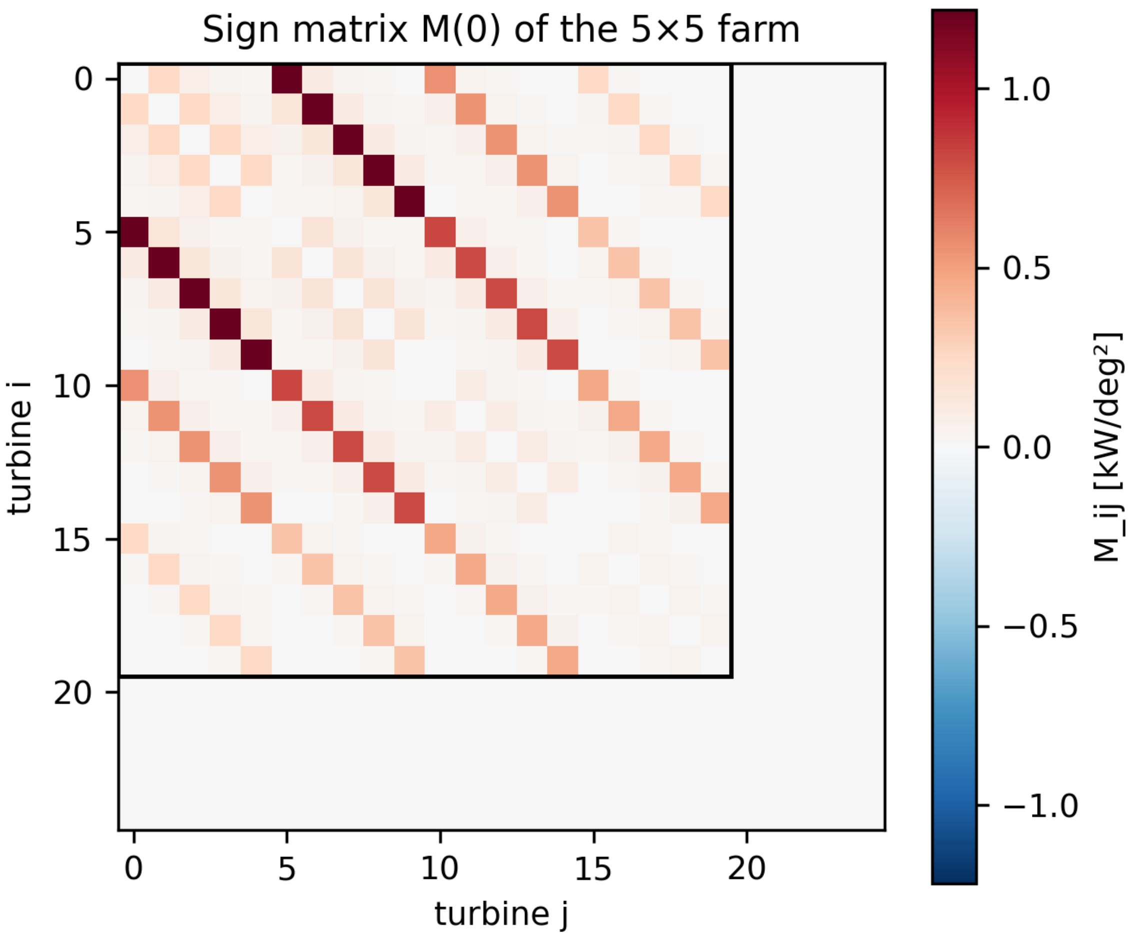

Figure 2: Sign matrix of the 5×5 farm at the origin: chain pairs dominate, lateral pairs are positive, the last row is decoupled, and thresholding at $\tau = 0.05\,\mathrm{kW\,deg}^{-2}$ recovers the wake DAG clusters (first four rows vs last row).

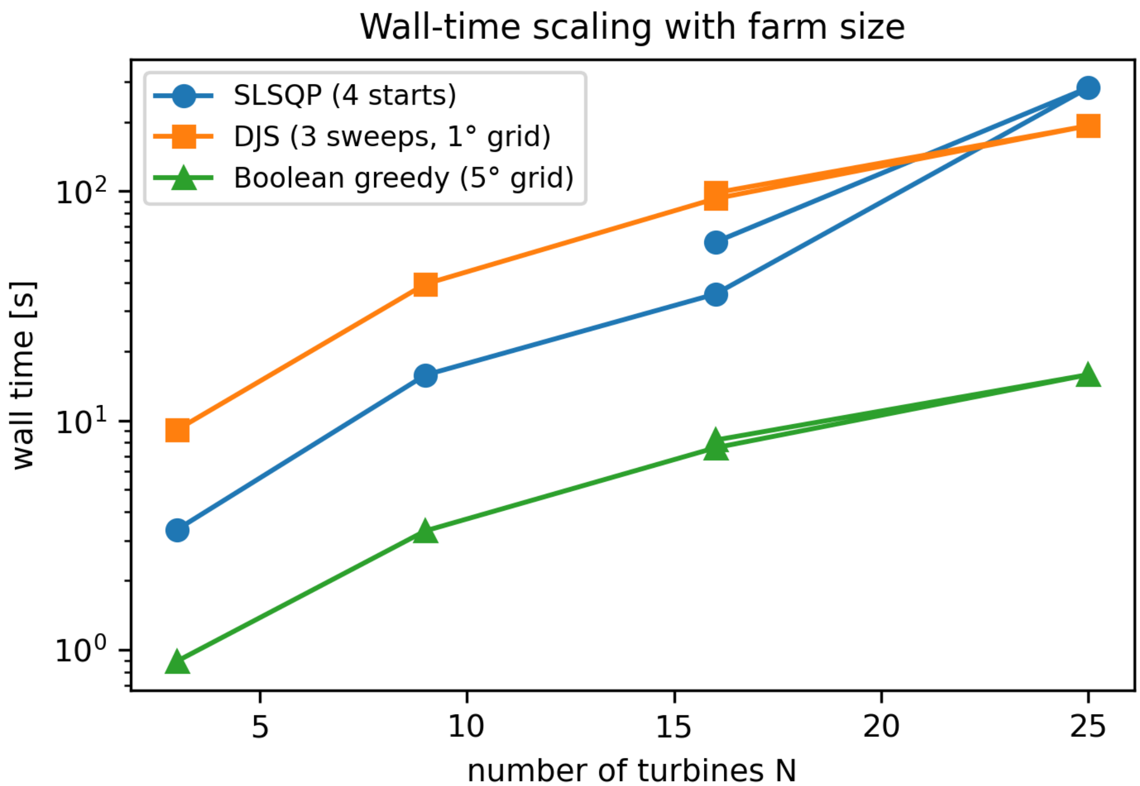

Figure 3: Wall-time scaling with farm size: Boolean greedy (5-degree grid) and DJS grow gently, SLSQP grows like the optimization workload it is.

5 The bounded-interaction certificate

Theorem 2 bounds the gap of any coordinate-sweep greedy method by the interaction energy. We now evaluate the bound on fresh, previously unseen layouts: 12 random farms ($N = 6$–9 turbines, wind directions 240–300°, seed 42), Boolean greedy on a 5° grid versus multi-start SLSQP, and $\bar{M}_{ij}$ sampled at the origin plus four random points of the box. Measured gaps: **mean 0.103 %, max 0.477 %** (one case: greedy 0.009 % *above* SLSQP, a multi-start artifact). Certificate: bound values span 0.12–7.05 % of farm power, in every case above the measured gap (Fig. 4): conservative by 1–2 orders of magnitude, as a certificate should be, and computable from N^2 model calls, cheaper than the optimization it certifies. To our knowledge this is the first guarantee of its kind for wake steering; it subsumes the empirical record (Boolean greedy $\leq 0.6\,\%$ (Stanley et al., 2022); per-row greedy 0.07 % in our companion platform) under one mechanism: greedy fails only through interaction energy, and interaction energy is what the sign matrix measures.

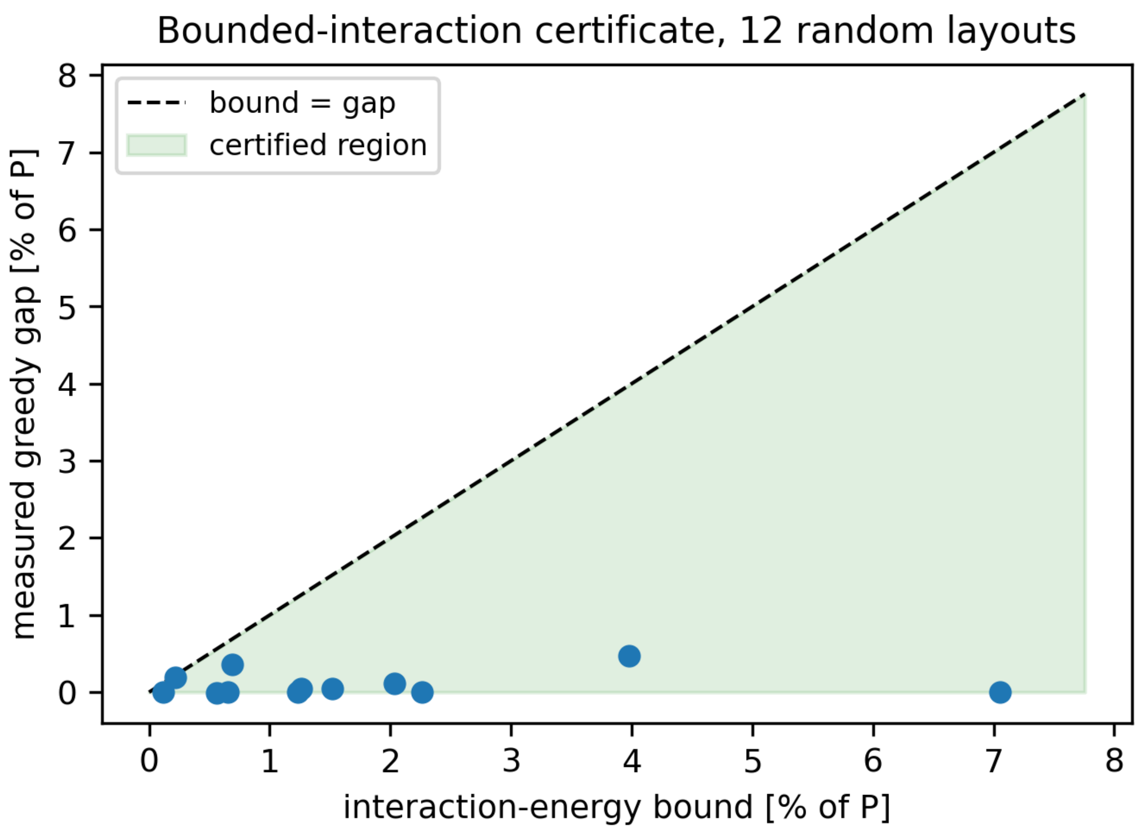

Figure 4: The certificate holds on all 12 random layouts: every measured greedy gap (mean 0.103 %, max 0.477 %) lies below the sampled interaction-energy bound (0.12–7.05 % of farm power).

6 Discussion and outlook

- **Loads:** yaw rate limits and fatigue constraints enter as per-turbine box constraints, which the decoupled sweeps handle without modification (they are per-coordinate).
- **Real-time:** two sweeps of $N \times 31$ evaluations at FLORIS speeds fit a 20-s control update cadence for tens of turbines; DJS is a drop-in replacement for serial-refine with a guarantee.
- **AEP:** the decoupling law holds at AEP optima (companion paper, Table 2), so DJS transfers to energy objectives unchanged.
- **RL:** the sign matrix is a natural communication-graph/credit-assignment prior for multi-agent RL (future work with our PPO pipeline).
- **Experimental path:** the DJS/certificate predictions are wind-tunnel measurable with torque-instrumented model turbines; the protocol (blocked pairs, 180-s averages, pre-specified thresholds) is in the companion paper's Appendix D. The certificate's premise (small interaction energy) is the quantity E2 of that protocol.

7 Conclusions

We have shown that the interaction structure of the companion paper is not just an explanation; it is a workable algorithm stack. Decoupled Jacobi sweeps exploit the near-diagonal Hessian at optima to reach SLSQP-level solutions in 2–3 parallel per-turbine passes; sign-matrix thresholding delivers certified decentralized clusterings that recover the centralized optimum at a fraction of the cost; and the interaction-energy certificate finally gives greedy methods the guarantee their 15-year empirical record deserves. All three tools rest on one structural object, the mixed-partial matrix, computed once per wind condition, and all three are stated against explicit failure boundaries (interior optima, sampled certificate, model class). The stack is the first wake-steering optimizer whose speed has a mechanism and whose gap has a bound.

A Reproduction

`exp_djs.py` (DJS benchmark table; 5×5 clustering), `exp_experiments2.py` (12-layout certificate benchmark; 5×5 sign matrix; wall-time scaling), `exp_traces_fix.py` (Table 1 gaps and times), `make_figures2.py`. FLORIS 4.6.6, `default_inputs.yaml`, $8\,\mathrm{m\,s^{-1}}$, TI 0.06, wd 270°, box $[0°, 30°]$. Finite differences central, $h = 5°$. Evaluations at simultaneous extreme yaw ($\geq 25°$ on many turbines) produce negative-velocity warnings in FLORIS and are excluded from claims. The certificate benchmark's random layouts are seed-42 draws, reproducible exactly.

Acknowledgements

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